Quantifying Ocular Confounds in EEG Decoding Without an Eye Tracker

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Abstract—Eye movements, blinks, and the corneo-retinal standing potential leak into scalp electroencephalography (EEG) and co-vary with task structure, so a classifier credited with decoding a cognitive state may instead be partly reading the eyes. In health applications this is not a bookkeeping matter: a communication brain–computer interface (BCI) that is covertly gaze-driven fails precisely for the patients with oculomotor impairment it is meant to serve. Existing defenses either remove ocular signal or model it within a single eye-tracked recording; none audits a completed decoding dataset that has no eye-tracker. We study a candidate auditing instrument. A frozen linear EEG-to-gaze readout, trained where synchronized gaze exists, induces a gaze-predictive EEG subspace (the orthogonal projection onto the span of its weights). The Ocular Predictability Index (OPI) is the cross-validated area under the curve (AUC) of a paradigm-matched decoder applied to that projection, minus a within-subject label-permutation null; it measures label predictability within the projection, not a fraction of the original effect and not a causal attribution. We validate against ground truth in a physically grounded EEG-plus-gaze simulator, and add the baselines and diagnostics needed to interpret it. Against a rank-matched random projection and a bipolar electrooculography (EOG) proxy, the gaze-trained subspace is distinguished only in the fixational regime (OPI 0.396 vs. 0.149 and 0.144 at 0.25°); at 2° and above all projections are equivalent within confidence intervals because the artifact dominates. It stays near-null on neural controls (OPI 0.007, 17% significant) where the random projection does not (0.072, 92%). Reporting reconstruction per ocular output rather than as one aggregate reverses the reliability verdict at the fixational extreme (horizontal-gaze $R^2 = 0.47$ vs. aggregate 0.66). We therefore present this as a simulation-validated proof of concept with a preliminary real-EEG illustration, not a calibrated transferable instrument: the decisive test, training the readout on real synchronized EEG–gaze data and transferring to independent participants, remains to be done.

Index Terms—EEG decoding, ocular artifact, eye movements, construct validity, confound auditing, brain–computer interface, subspace projection

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I. Introduction

A decoding result is only as trustworthy as the signal it rests on. In EEG, a recurring threat to construct validity is that a classifier credited with reading a cognitive state (attention, working-memory load, target detection, the meaning of a read word) is partly reading the eyes. The eye is an electrical dipole, cornea-positive and retina-negative, with a standing potential of roughly 0.4–1.0 mV [1]; its rotations, together with blinks and the broadband saccadic spike potential at saccade onset [2], project to frontal and periocular electrodes with amplitudes that dwarf cortical signals. The field of an eye movement depends on each electrode's position relative to the eyes: its sign reverses across the midline and its magnitude falls with distance, so contamination is montage- and artifact-specific. Because where and when a person looks is rarely independent of the experimental condition, this ocular activity is condition-correlated: it can be decoded in lockstep with the cognitive label and inflate apparent accuracy — the machine-learning failure mode of shortcut learning [3].

The clinical stakes are concrete. A communication BCI validated on healthy participants whose decodable signal is substantially ocular will degrade or fail in amyotrophic lateral sclerosis and other conditions impairing oculomotor control — the population the device targets. An EEG biomarker of attention or workload that is partly gaze-driven tracks looking behaviour rather than the neural state it is billed as measuring, and will not transfer across populations whose eye-movement statistics differ. Because eyes and task are correlated by design in most paradigms, the confound survives group averaging and is invisible in accuracy alone.

The standard defenses are incomplete. Independent component analysis and regression remove ocular components [4], but removal is imperfect and leaves a residual whose decodable content is unknown; overweighting the saccadic spike before ICA improves this but still targets removal, not measurement [5]. Deconvolution frameworks rigorously model ocular and overlap effects, but in practice within a single, carefully co-registered, eye-tracked recording [6]. None answers the question an author or reviewer faces about a finished result recorded with no eye-tracker: how much of this decodable signal could be ocular?

The concern is not hypothetical. Mostert et al. [7] showed that a visual working-memory "representation" could be partly explained by systematic eye movements, and constructed a bespoke, within-experiment control. Rotaru et al. [8] found that EEG decoders of the spatial focus of auditory attention overfit to eye-gaze and per-trial fingerprints, dropping to chance once gaze was controlled. More broadly, the field increasingly rewards auditing its methods and benchmarks for inflated inference [9]–[11].

This paper defines such an audit and, as importantly, establishes what it can and cannot currently support. Our contributions are: (i) a frozen linear gaze-predictive subspace operator and an Ocular Predictability Index (OPI) with a within-subject permutation null and a full-EEG ceiling; (ii) a comparison against simple alternatives — a rank-matched random projection, a bipolar EOG proxy, and a frontal-channel audit — which localises the instrument's advantage to the fixational regime and shows it has none at large amplitudes; (iii) a reliability gate reported per ocular output, strictly more conservative than the aggregate it replaces; and (iv) a subspace-overlap diagnostic for the residual failure mode of absorbing genuine neural signal, with an account of why the Haufe activation pattern is the wrong object here.

II. Methods

A. Design and rationale

The instrument is deliberately linear. Although a deep network is one way to learn an EEG-to-gaze readout where synchronized eye-tracking is available [12], [13], the object that would be frozen and shipped is a linear readout and the orthogonal projection onto the span of its weights, so we implement and validate exactly that. Because ground truth is available only when gaze is recorded or simulated, the primary validation uses a forward-model simulator with known sources. The auditor uses every channel a dataset provides; we report a full 64-channel 10–10 montage and treat the 30-channel ERP-CORE set [14] as an explicitly specified subset used to characterise how channel count affects reliability.

B. The frozen gaze-predictive operator

Let $\mathbf{X}_{\mathrm{src}} \in \mathbb{R}^{n\times p}$ be source EEG (trials $\times$ channels) for which ocular ground truth $\mathbf{Y}_{\mathrm{src}} \in \mathbb{R}^{n\times q}$ is available (q outputs: horizontal and vertical gaze, a saccade channel, and a blink indicator). A ridge readout $\mathbf{W} = \arg\min_{\mathbf{W}} \|\mathbf{Y}_{\mathrm{src}} - \mathbf{X}_{\mathrm{src}}\mathbf{W}\|_F^2 + \lambda\|\mathbf{W}\|_F^2$ maps EEG to ocular state. The gaze-predictive subspace is the column space of $\mathbf{W}$, and the locked operator is the orthogonal projection onto it,

$$\mathbf{P} = \mathbf{U}\mathbf{U}^{\top}, \qquad \mathbf{U} = \text{left singular vectors of } \mathbf{W}, \tag{1}$$

a symmetric idempotent $p \times p$ matrix. Projecting target EEG, $\mathbf{X} \mapsto \mathbf{X}\mathbf{P}$, retains only the component lying in the readout's span. We name this subspace by what it demonstrably is — the set of EEG directions from which gaze is linearly predictable — and not "ocular," because an EEG-to-gaze readout may also capture genuine cortical activity associated with eye movements, visual attention, motor preparation, or task structure. Establishing specificity to ocular electrical contamination would require an independent manipulation we do not perform; this identification problem is the principal limitation of the approach (Section IV-A).

For a target with a different montage, the readout is refit on source data restricted to that montage (no periocular interpolation); the operator is frozen per montage and applied unchanged to every target. Montage matching by shared channel names is necessary but not sufficient for transfer: reference scheme, electrode coordinates, cap placement, hardware, filtering, and participant anatomy all modulate the ocular scalp projection and are not controlled by name matching.

C. The Ocular Predictability Index

A paradigm-matched cognitive decoder (a ridge linear classifier) is trained and tested under strictly subject-wise cross-validation on the projected data $\mathbf{X}\mathbf{P}$, yielding a macro-averaged AUC. The OPI subtracts a within-subject label-permutation null (labels shuffled within each subject, $B = 200$):

$$\mathrm{OPI} = \mathrm{AUC}_{\mathrm{proj}} - \overline{\mathrm{AUC}}_{\mathrm{null}}, \tag{2}$$

reported alongside the full-EEG AUC as a ceiling and a permutation p-value.

The OPI must be read for exactly what it is: how much label information survives in the gaze-predictive projection. It is a difference of AUCs, not a proportion, and it is not a fraction of the original decoding effect. A high OPI does not establish that ocular artifact caused the observed decoding performance — genuine gaze-correlated neural activity, task structure, or any other signal aligned with the projection would produce the same reading. A low OPI is likewise not a clean bill of health: if a source-trained readout fails to capture the target's ocular artifacts, absence of detection reflects failed transfer rather than absence of contamination.

D. Baselines

To establish whether gaze training does work that a simpler construction would not, we compute the OPI under three comparators on identical data, differing only in the subspace projected onto: a random subspace matched in rank; an EOG proxy from bipolar channel arithmetic alone (F8 − F7, (Fp1 + Fp2)/2); and a frontal-channel audit, the coordinate subspace of the 17 frontopolar, anterior-frontal and frontal electrodes.

E. Reliability gate and bias envelope

Reliability is the subject-wise cross-validated gaze-reconstruction of the readout restricted to the target montage, computed on source data (leave-one-subject-out over source subjects), never on the target. Because the four ocular outputs are heterogeneous, we report them separately rather than as one aggregate: R^2 for the three continuous targets (horizontal gaze, vertical gaze, saccade channel) and AUC for the binary blink indicator, for which an R^2 is not interpretable. The gate is applied to the output that the audited contrast depends on — horizontal gaze for lateralized paradigms.

A no-eye-tracker target inherits the score of its matched source montage; where none matches, the gate returns inconclusive. This establishes competence within the source distribution only: it does not demonstrate reliability in an unseen target dataset, which would require held-out real participants with measured gaze. Thresholds are fit on one montage grid and evaluated on a held-out grid (Section III-D), reported to two significant figures, and should be treated as preliminary. The bias envelope is measured: where ground-truth gaze exists, we compare the auditor's OPI to the OPI computed directly from true gaze across the saccade-amplitude range and report the signed deviation as a two-sided margin.

F. The subspace-overlap diagnostic

The instrument's central risk is that the projection absorbs genuine neural activity that happens to be gaze-correlated, inflating the index. For a target whose cognitive decoder has discriminative direction $\mathbf{t}$, define the overlap

$$s = \frac{\|\mathbf{P}\mathbf{t}\|}{\|\mathbf{t}\|} = \cos\theta, \tag{3}$$

the cosine of the principal angle between $\mathbf{t}$ and the gaze-predictive subspace. This is a geometric diagnostic, not a bound: converting s into decodable leakage depends on the decoder, noise covariance, and signal amplitude, and we make no closed-form claim. We report s per audit and show empirically that leakage increases with s over most of its range.

A reasonable objection is that a discriminative weight vector is not a valid activation pattern, and that the Haufe transform $\mathbf{a} = \mathbf{\Sigma}_X\mathbf{t}$ should be used instead [15]. We compute both and report the comparison, because in this setting the transform is actively counter-productive: $\mathbf{\Sigma}_X$ is dominated by the very ocular variance the diagnostic is meant to be separate from, so $\mathbf{a}$ points into the gaze-predictive subspace almost regardless of $\mathbf{t}$ (Section III-E). The object that determines what the decoder actually reads — and therefore what can leak — is the discriminative direction, and that is what we use.

G. Simulator and real data

Ground-truth validation uses a forward-model simulator on a full 64-channel 10–10 montage. Per-trial channel patterns are sums of physically grounded source topographies: a horizontal corneo-retinal field ($\approx 16\,\mu$V per degree), a vertical/blink frontopolar field, a focal periocular saccadic spike scaling with saccade size [2], an $\approx 80\,\mu$V frontopolar blink, and a lateralized posterior N2pc-like neural component ($\approx 1.8\,\mu$V) [16], on spatially correlated background noise. An artifact is coupled to the cognitive label with tunable strength. Each trial is a single channel-space feature vector; this omits the temporal morphology distinguishing sustained corneo-retinal shifts, blink waveforms, saccadic spikes, and event-related potentials, and is a substantive simplification. The readout is trained on 20 source subjects $\times$ 200 trials at 8° saccades; each evaluation cell is 12 subjects $\times$ 120 trials over 12 fixed-seed studies, with ridge λ=1 fixed, leave-one-subject-out macro AUC, and a B=200 within-subject label-permutation null. Uncertainties are 95% confidence intervals over the 12 studies, except in the leakage sweep of Section III-E, which uses 10 subjects over 8 studies. Generative equations and parameters are in the released code.

For real data we use PhysioNet eegmmidb (all 64 channels, BCI2000, 160 Hz; 8 subjects) [17], [18]. Because eegmmidb has no eye-tracker, no gaze-trained operator can be built for it; the subspace is instead estimated from the eyes-open baseline run of four source subjects and applied to four held-out target subjects' motor runs. This is an illustrative proxy, not a test of the proposed instrument, and the sample is too small for stable subject-level estimates.

H. Implementation

The auditor, simulator, and analyses are implemented in R (CPU-only). AUC is computed by the rank statistic and cross-checked against the pROC package to $< 10^{-6}$.

III. Results

A. The operator behaves as specified

Unit tests confirm the rank-based AUC matches pROC to $< 10^{-6}$; the projector of (1) is symmetric, idempotent, and correct-rank; and end-to-end the OPI is large when the cognitive signal lies in the projection and null when it lies outside. The trained readouts had rank 4 on the simulator and rank 2 on real eegmmidb.

B. Sensitivity, specificity, and what the baselines show

On the full 64-channel montage the OPI separates condition-correlated ocular contamination from neural signal, and does so for two distinct artifact types: saccade-coupled ocular gives OPI $= 0.429 \pm 0.005$ with projected AUC at the full-EEG ceiling, and a condition-correlated eye-blink artifact gives 0.198 ± 0.008, both significant in 100% of studies. Negative controls are near-null: motor

TABLE I
OPI by projection (mean over 12 simulated studies, full 64-channel montage; 95% CIs ≤ 0.033 throughout, given per cell in the text). Positive controls at 8°; the fixational row is the discriminating one.

Scenario	gaze k=4	random k=4	EOG k=2	frontal k=17
Saccade-coupled, 8°	0.429	0.428	0.429	0.429
Saccade-coupled, 0.25°[a]	0.396	0.149	0.144	0.401
Blink-coupled	0.198	0.199	0.175	0.195
Covert attention[b]	0.007	0.072	0.008	−0.004
Motor imagery[b]	0.031	0.072	0.009	0.004

[a] Fixational regime: the only positive-control row on which the projections differ.
[b] Neural negative controls; % of studies significant is 17 (gaze), 92 (random), 17 (EOG), 8 (frontal) for covert attention.

imagery 0.031±0.008, covert attention 0.007±0.012, label-shuffle 0.007, neutral −0.003. The covert-attention case is the informative one: the contrast is decodable from full EEG (ceiling AUC 0.79) yet not from the projection.

The baselines qualify this substantially (Table I). At the 8° saccade amplitude used for the headline battery, every projection — including a random rank-4 subspace — recovers OPI ≈ 0.43. That result carries no information about the merit of gaze training: at 8° the ocular signal ($\approx 128\,\mu$V) so dominates the neural source ($1.8\,\mu$V) and noise ($3\,\mu$V) that almost any subspace retains it. The methods separate only as amplitude falls. At 0.25° — the fixational regime that survives eye-movement rejection, and the regime the audit is actually for — the gaze-trained subspace gives 0.396 ± 0.015 against 0.149 ± 0.033 for the rank-matched random projection and 0.144 ± 0.013 for the bipolar EOG proxy.

Specificity separates the comparators differently. Against the covert-attention neural control, the random projection returns 0.072 ± 0.019 and is significant in 92% of studies — a false ocular attribution — where the gaze-trained subspace returns 0.007 (17%). The EOG proxy is likewise specific (0.008, 17%) but insensitive in the fixational regime. The frontal-channel audit matches the gaze-trained operator on both axes in this simulator (0.401 at 0.25°; −0.004 on covert attention), achieving with rank 17 what the gaze-trained operator achieves with rank 4. We therefore do not claim superiority over frontal-channel auditing on this evidence. Notably, the simulator contains no frontal neural generator, so it cannot discriminate the two; the real-data proxy below is where the frontal approach visibly fails.

The index is graded rather than binary: as the gaze–label coupling ρ increases from 0 to 1 in steps of 0.2, the OPI rises monotonically and near-linearly (−0.013, 0.100, 0.202, 0.303, 0.404, 0.499), with the projected AUC tracking from chance to 1.0.

TABLE II
Per-output reconstruction (R^2 for continuous targets, AUC for the binary blink indicator). The aggregate hides the horizontal-gaze collapse.

Amp. (deg)	gaze_x R^2	gaze_y R^2	sacc. R^2	blink AUC	aggr. R^2
Full montage (64 ch)					
0.25	0.474	0.880	0.469	1.000	0.664
1.0	0.959	0.872	0.948	1.000	0.899
12.0	0.999	0.633	0.988	0.979	0.755
ERP-CORE subset (30 ch)					
0.25	0.353	0.761	0.349	0.993	0.497
1.0	0.923	0.754	0.913	0.991	0.775
12.0	0.999	0.505	0.988	0.934	0.679

Means over 12 studies; SDs ≤ 0.03 throughout.

C. Per-output reliability changes the verdict

Aggregating reconstruction across the four heterogeneous ocular outputs masks the failure that matters (Table II). Blink detection is near-perfect everywhere (AUC $\geq$ 0.93) and vertical gaze is reconstructed well even at small amplitudes, so the aggregate stays high; but horizontal gaze — the dimension a lateralized paradigm depends on — collapses at the fixational extreme, to $R^2 = 0.474$ on the full montage and 0.353 on the 30-channel subset. Under the aggregate score the full montage at 0.25° passes the confirmatory threshold (0.664); under the horizontal-gaze score it does not (0.474). Reporting per output is thus strictly more conservative and reverses a verdict, which is the behaviour an audit instrument should have.

D. Bias envelope and held-out thresholds

Comparing the auditor's OPI to the OPI from ground-truth gaze across the saccade-amplitude range, the bias is within ±0.01 from 0.5° upward on both montages. Only at the fixational extreme does it grow, to −0.034 (full) and −0.056 (30-channel), and in both cases the auditor understates, the conservative direction. No over-statement was observed within the conditions swept; this envelope covers simulated horizontal saccades only and does not generalise to real blinks, microsaccades, smooth pursuit, varying references, or other acquisition systems.

Thresholds were previously fit and evaluated on the same grid. Refitting on the full-64 grid alone and applying the result unchanged to the held-out 30-channel grid, the confirmatory threshold is $R^2 \approx 0.66$; six of seven held-out cells are tiered confirmatory and all six have $|\text{bias}| \leq 0.05$, while the 0.25° cell is correctly excluded. Under this honest calibration the confirmatory and exploratory thresholds coincide, so the evidence supports a two-tier gate (confirmatory / inconclusive), not the three tiers previously reported. The value is preliminary and specific to this simulator.

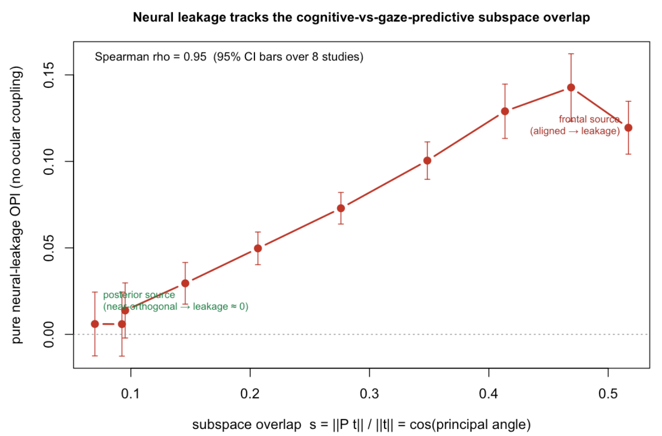

Fig. 1. Neural leakage versus the overlap $s = \cos\theta$ between the cognitive discriminative direction and the gaze-predictive subspace. Leakage increases with s ($\rho = 0.95$; not strictly monotone at the highest overlap) and is ≈ 0 for a posterior (near-orthogonal) source. Bars are 95% CIs over 8 studies.

E. Neural leakage tracks the overlap angle

Sweeping a lateralized neural source from posterior to frontal with no ocular coupling — so any positive OPI is pure leakage — leakage increases with the overlap s of (3) (Fig. 1, Spearman $\rho = 0.95$), from $+0.006 \pm 0.018$ at near-orthogonality to a maximum of $+0.143 \pm 0.020$. The relation is not strictly monotone: at the most frontal source position, the highest overlap tested, leakage falls back to $+0.119 \pm 0.015$, so s orders the regime rather than predicting leakage exactly. The overlap computed on the decoder's discriminative direction ranks the conditions identically to the overlap computed on the true source topography ($\rho = 1.00$), so the empirical direction is an adequate stand-in for the generator here.

The Haufe-transformed activation pattern behaves differently, and instructively: its overlap with the gaze-predictive subspace saturates in the range 0.92–0.99 across every source position, and correlates negatively with leakage ($\rho = -0.54$). The reason is structural — the data covariance is dominated by ocular variance, so $\mathbf{\Sigma}_X \mathbf{t}$ is pulled into the gaze-predictive subspace whatever $\mathbf{t}$ is. The activation pattern is the right object for interpreting what generated a component; it is the wrong object for asking what a decoder reads, which is what governs leakage.

F. An illustrative real-EEG proxy audit

On real 64-channel eegmmidb, with a subspace frozen from four source subjects' eyes-open baseline and applied to four held-out target subjects, the left-versus-right imagery contrast is weakly decodable from full EEG (ceiling AUC 0.601) but not from the projection (projected AUC 0.501, OPI -0.023, $p = 0.998$). This is a plausible negative control rather than a measured one: without an eye-tracker we cannot verify that left- and right-hand imagery involve no differential gaze. Rest-versus-move is the cautionary case: the channel-proxy subspace retains frontal neural signal it cannot separate from ocular activity (ceiling 0.672, projected 0.615, OPI $+0.114$, $p = 0.002$) — a false ocular attribution. (Macro-averaging over few subjects raises the permutation null to ≈ 0.52, so a negative OPI means projected AUC below that null.) With eight participants these estimates are unstable and are offered as illustration only. Their value is negative evidence: they show what a proxy subspace does wrong, and thereby why a readout trained on real synchronized gaze — which this experiment does not implement — is the necessary next step.

IV. Discussion

The construct behaves as specified in simulation: sensitive to two distinct artifact types, near-null on neural and shuffled controls, graded in the coupling, conservative in its measured bias, and equipped with a gate that can return "inconclusive." The baselines mark out what the evidence does not support: gaze training earns its keep only in the fixational regime, and a 17-channel frontal audit matches the rank-4 operator throughout this simulator. What the comparison does establish is that subspace choice matters — a rank-matched random projection produces significant false ocular attributions on a purely neural contrast in 92% of studies.

Relation to prior work. The construct generalizes the within-experiment control of Mostert et al. [7] and the single-domain diagnosis of Rotaru et al. [8] toward a target-agnostic meter, complementing ocular removal [4], [5] and within-recording deconvolution [6]. Where confound-quantification in machine learning probes a measured nuisance [11], here the nuisance is inferred from EEG itself — the source of both the method's reach and its central ambiguity.

A. Limitations

The identification problem is primary. An EEG-to-gaze readout learns the EEG directions that predict gaze; these need not be exclusively corneo-retinal, blink, and saccadic contamination, and may include genuine cortical activity tied to eye movements, visual attention, motor preparation, or task structure. We therefore name the object a gaze-predictive subspace and read the OPI as label predictability within it. Demonstrating specificity to ocular electrical contamination requires an independent manipulation — for example, comparing minimally- and maximally-preprocessed versions of the same recording — which we have not performed.

The validation is simulation-based, and the simulator's linear, spatial-only generative structure is aligned with the linear spatial operator being tested, which limits what internal success demonstrates. Reliability is measured on source data; transfer to unseen target participants is assumed, not shown, and montage matching by channel name does not control reference, geometry, hardware, or anatomy. The thresholds are preliminary, support two tiers rather than three, and were calibrated within one simulator. The bias envelope covers simulated horizontal saccades only. The overlap score is a geometric diagnostic whose increasing relation to decodable leakage is empirical rather than derived, and is not strictly monotone at high overlap. The real-EEG experiment uses a channel proxy, not the proposed gaze-trained operator, on eight participants. Finally, the index addresses only the gaze-predictive subspace and will not detect non-ocular confounds such as repeated-stimulus fingerprints.

Accordingly we claim no calibrated, field-deployable instrument, and propose no cross-benchmark leaderboard on this evidence: paradigms differ too much in ocular behaviour, neural sources, montage, reference, and preprocessing for one frozen operator to be assumed valid across them.

Next steps. The decisive experiment remains to be run: train the readout on real synchronized EEG–gaze recordings [12], [13], measure gaze reconstruction in held-out real participants rather than inheriting a source score, and transfer the frozen operator to an independent dataset. Comparing minimally- against maximally-preprocessed versions of the same recordings would test ocular specificity directly. Because EEGEyeNet uses cued large saccades, the sub-degree regime — where our baselines locate the method's only advantage — must be anchored on reading microsaccades.

V. Conclusion

How much of an EEG decoding result could be the eyes rather than the mind is, for most published datasets, unanswerable because no eye-tracker was present. We have defined a frozen, montage-matched gaze-predictive subspace and an Ocular Predictability Index that makes the question operational, validated its behaviour against ground truth in simulation, and — by adding rank-matched baselines, per-output reliability, and an overlap diagnostic — delimited what that validation supports. Its failure modes are now measurable rather than rhetorical; the transferable instrument is not yet established, and real synchronized-gaze validation is the next step rather than a completed claim.

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References

[1] J. Malmivuo and R. Plonsey, Bioelectromagnetism: Principles and Applications of Bioelectric and Biomagnetic Fields. New York: Oxford University Press, 1995.

[2] A. S. Keren, S. Yuval-Greenberg, and L. Y. Deouell, "Saccadic spike potentials in gamma-band EEG: Characterization, detection and suppression," NeuroImage, vol. 49, no. 3, pp. 2248–2263, 2010.

[3] R. Geirhos, J.-H. Jacobsen, C. Michaelis, R. Zemel, W. Brendel, M. Bethge, and F. A. Wichmann, "Shortcut learning in deep neural networks," Nature Machine Intelligence, vol. 2, no. 11, pp. 665–673, 2020.

[4] T.-P. Jung, S. Makeig, C. Humphries, T.-W. Lee, M. J. McKeown, V. Iragui, and T. J. Sejnowski, "Removing electroencephalographic artifacts by blind source separation," Psychophysiology, vol. 37, no. 2, pp. 163–178, 2000.

[5] O. Dimigen, "Optimizing the ICA-based removal of ocular EEG artifacts from free viewing experiments," NeuroImage, vol. 207, p. 116117, 2020.

[6] B. V. Ehinger and O. Dimigen, "Unfold: an integrated toolbox for overlap correction, non-linear modeling, and regression-based EEG analysis," PeerJ, vol. 7, p. e7838, 2019.

[7] P. Mostert, A. M. Albers, L. Brinkman, L. Todorova, P. Kok, and F. P. de Lange, "Eye movement-related confounds in neural decoding of visual working memory representations," eNeuro, vol. 5, no. 4, pp. ENEURO.0401–17.2018, 2018.

[8] I. Rotaru, S. Geirnaert, N. Heintz, I. Van de Ryck, A. Bertrand, and T. Francart, "What are we really decoding? unveiling biases in EEG-based decoding of the spatial focus of auditory attention," Journal of Neural Engineering, vol. 21, no. 1, p. 016017, 2024.

[9] S. Kapoor and A. Narayanan, "Leakage and the reproducibility crisis in machine-learning-based science," Patterns, vol. 4, no. 9, p. 100804, 2023.

[10] G. Brookshire, J. Kasper, N. M. Blauch, Y. C. Wu, R. Glatt, D. A. Merrill, S. Gerrol, K. J. Yoder, C. Quirk, and C. Lucero, "Data leakage in deep learning studies of translational EEG," Frontiers in Neuroscience, vol. 18, p. 1373515, 2024.

[11] E. Chaibub Neto, "Using permutations to detect, quantify and correct for confounding in machine learning predictions," arXiv preprint arXiv:1805.07465, 2018.

[12] A. Kastrati, M. B. Płomecka, D. Pascual, L. Wolf, V. Gillioz, R. Wattenhofer, and N. Langer, "EEGEyeNet: A simultaneous electroencephalography and eye-tracking dataset and benchmark for eye movement prediction," in Advances in Neural Information Processing Systems (NeurIPS), Track on Datasets and Benchmarks, 2021, arXiv:2111.05100.

[13] N. Hollenstein, J. Rotsztejn, M. Troendle, A. Pedroni, C. Zhang, and N. Langer, "ZuCo, a simultaneous EEG and eye-tracking resource for natural sentence reading," Scientific Data, vol. 5, p. 180291, 2018.

[14] E. S. Kappenman, J. L. Farrens, W. Zhang, A. X. Stewart, and S. J. Luck, "ERP CORE: An open resource for human event-related potential research," NeuroImage, vol. 225, p. 117465, 2021.

[15] S. Haufe, F. Meinecke, K. Görgen, S. Dähne, J.-D. Haynes, B. Blankertz, and F. Bießmann, "On the interpretation of weight vectors of linear models in multivariate neuroimaging," NeuroImage, vol. 87, pp. 96–110, 2014.

[16] S. J. Luck and S. A. Hillyard, "Electrophysiological correlates of feature analysis during visual search," Psychophysiology, vol. 31, no. 3, pp. 291–308, 1994.

[17] G. Schalk, D. J. McFarland, T. Hinterberger, N. Birbaumer, and J. R. Wolpaw, "BCI2000: A general-purpose brain-computer interface (BCI) system," IEEE Transactions on Biomedical Engineering, vol. 51, no. 6, pp. 1034–1043, 2004.

[18] A. L. Goldberger, L. A. N. Amaral, L. Glass, J. M. Hausdorff, P. C. Ivanov, R. G. Mark, J. E. Mietus, G. B. Moody, C.-K. Peng, and H. E. Stanley, "PhysioBank, PhysioToolkit, and PhysioNet: Components of a new research resource for complex physiologic signals," Circulation, vol. 101, no. 23, pp. e215–e220, 2000.